

Leonidas Papanastasiou

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Software Engineer with 6+ years of experience building scalable cloud-native systems across banking, capital markets, energy, and supply chain sectors. Proven track record building high-performance applications using **Python (Django, FastAPI, Flask)**, **React** and **blockchain technologies (Hyperledger, Solidity)**, serving 5,000+ users. Specialized in trading and **digital-asset platforms**, from quantitative hedge fund infrastructure to regulated banking, as well as real-time data pipelines and low-latency, high-throughput systems.

PROFESSIONAL EXPERIENCE

UpcoMinds | Athens, Greece | Consultant - Software Engineer | 06/2026 – Present

- Engineered a major Greek bank's retail **digital-asset** investment platform, bringing OMS experience into regulated banking: 5 async **Python (FastAPI)** services with 80+ REST endpoints on **SQLAlchemy** and **PostgreSQL/TimescaleDB**, covering onboarding, order lifecycle, positions and regulatory compliance.
- Built the bank's back-office control plane for operations, configuration, IT and AML teams: a per-asset **trading kill switch**, product onboarding, forced liquidations and a bank-wide exposure view, all behind **maker-checker (four-eyes)** approvals and **Microsoft Entra ID** role-based access control.
- Integrated with the **core banking** system to reserve customer funds before each trade, with self-healing end-of-day reconciliation of settlements, expired orders and timed-out fund holds on **Azure Container Apps Jobs**, using Postgres advisory locks for single-instance execution.
- Streamed real-time **WebSocket** market data through **Azure Web PubSub** into **TimescaleDB**, and hardened the platform with idempotency keys, circuit breakers, an append-only audit trail and **OpenTelemetry** tracing, shipped via **Azure DevOps** CI/CD, **Docker** and **Bicep** with 1,900+ tests behind an 85% coverage gate.

Quantitative Hedge Fund | New York, US | Consultant - Quantitative Software Engineer | 06/2025 – 03/2026

- Collaborated with trading, operations, and technology teams to design scalable internal software solutions, using **Python Django**, **AWS (EC2, S3)**, **PostgreSQL** and **Redis** for caching, aligning quantitative strategies with technical infrastructure.
- Enhanced the firm's Order Management System (OMS), improving CI/CD pipelines with **GitHub Actions**, introducing **Celery** for background task processing and optimizing system performance, execution latency and reliability across high-frequency trading pipelines.
- Identified and resolved key system bottlenecks using **Prometheus** and **Grafana** monitoring, along with **cProfile** profiler runs, boosting application availability and throughput in high-volume trading environments.

Wings ICT Solutions | Athens, Greece | Blockchain & Data Engineer | 04/2023 – 05/2025

- Architected scalable private blockchain network systems, using **Hyperledger Besu**, implementing custom smart contracts (**Solidity**) and permissioned consensus mechanisms for enhanced supply chain transparency and integrity.
- Implemented a **Python Flask**-based RESTful microservices to expose 20+ endpoints for blockchain interaction layers, enabling faster retrieval and analytics.
- Orchestrated serverless data ingestion using **Azure Functions** and **API Management**, unifying 7 distinct sources into a single **MongoDB** datastore, resulting in a 50% reduction in data latency and reducing infrastructure costs by 25%.

Evolution Projects Plus | Athens, Greece | Full Stack Engineer / Product Manager | 09/2020 – 04/2023

- Delivered a **Python Django**-based Waste Data Management System using **MySQL** and **ML**, forecasting waste volumes with 92% accuracy using **scikit-learn**, adopted by 45+ municipalities.

- Led end-to-end product development of award-winning enterprise platform: a **Python FastAPI**, **ReactJS** and **PostgreSQL**-based Dynamic Workflow Management System for DEDA S.A., scaling to 5,000+ unique users and managing 10,000+ user applications.
- Built multi-platform energy complaint portals (RAE) and real-time geospatial fault systems (Mytilineos) using **PostGIS** and **Leaflet.js**, along with **Python FastAPI**, **ReactJS**, **PostgreSQL** and **Redis**, handling 2,000+ field tickets.

Vision TV Entertainment | Athens, Greece | Cofounder & Lead Engineer | 06/2016 – 04/2017

- Co-founded and engineered an end-to-end OTT content delivery platform combining **Python** with custom **Linux** distribution on a **Raspberry Pi** edge devices to modernize TV distribution for the Greek market.
- Responsible for system architecture and development across backend services and device software, focused on reliability and performance in bandwidth-constrained environments.
- Integrated the platform with one of Greece's largest telecom networks, collaborating on technical integration and pilot rollout.

SKILLS

Languages: Python, SQL, JavaScript/TypeScript, Go, Rust, C/C++

Frameworks: Django, Flask, FastAPI, SQLAlchemy, Pydantic, ReactJS, Next.js, Flutter

Blockchain & Digital Assets: Hyperledger Besu, Ethereum (Solidity), Solana (Rust), Hardhat, Foundry

Data & ML: Pandas, Numpy, Scikit-learn, PyTorch, TensorFlow, Airflow

Tools: Git, PostgreSQL, TimescaleDB, MySQL, MongoDB, Redis, Prometheus, Grafana, OpenTelemetry, Celery

Foreign Languages: Greek (Native) | English (C2) | German (B1)

Cloud & Infrastructure: AWS (EC2, S3), Azure (Container Apps, Functions, Web PubSub, Entra ID, Monitor), Docker, Azure DevOps, GitHub Actions, Bicep

EDUCATION

MSc in Data Science & Information Technologies

National and Kapodistrian University of Athens, Greece

- Specialization: Big Data & Artificial Intelligence

BSc in Informatics and Telecommunications

National and Kapodistrian University of Athens, Greece

- Top 3 freshman student award
- Upper Second Class Honors (2:1)
- Member of Student Council

ACCOMPLISHMENTS & CERTIFICATIONS

- 2025 | Specialization in Machine Learning for Trading | Coursera / Google Cloud
- 2025 | Quantitative Finance & Algorithmic Trading in Python | Udemy
- 2024 | Ultimate Rust Crash Course | Udemy
- 2024 | AI Engineering Professional Certificate | Coursera / IBM
- 2022 | "Best Information System & IoT Application" Award for Dynamic Workflow Management System (DEDA S.A.)
- 2017 | Top3 Startup | Athens Startup Weekend
- 2016 | Speaker | 18th InfoCom World Conference | Presented my cofounded startup, Vision TV Entertainment
- 2008 | 3rd Place | Hellenic Society of Mathematics "Thalis" Contest